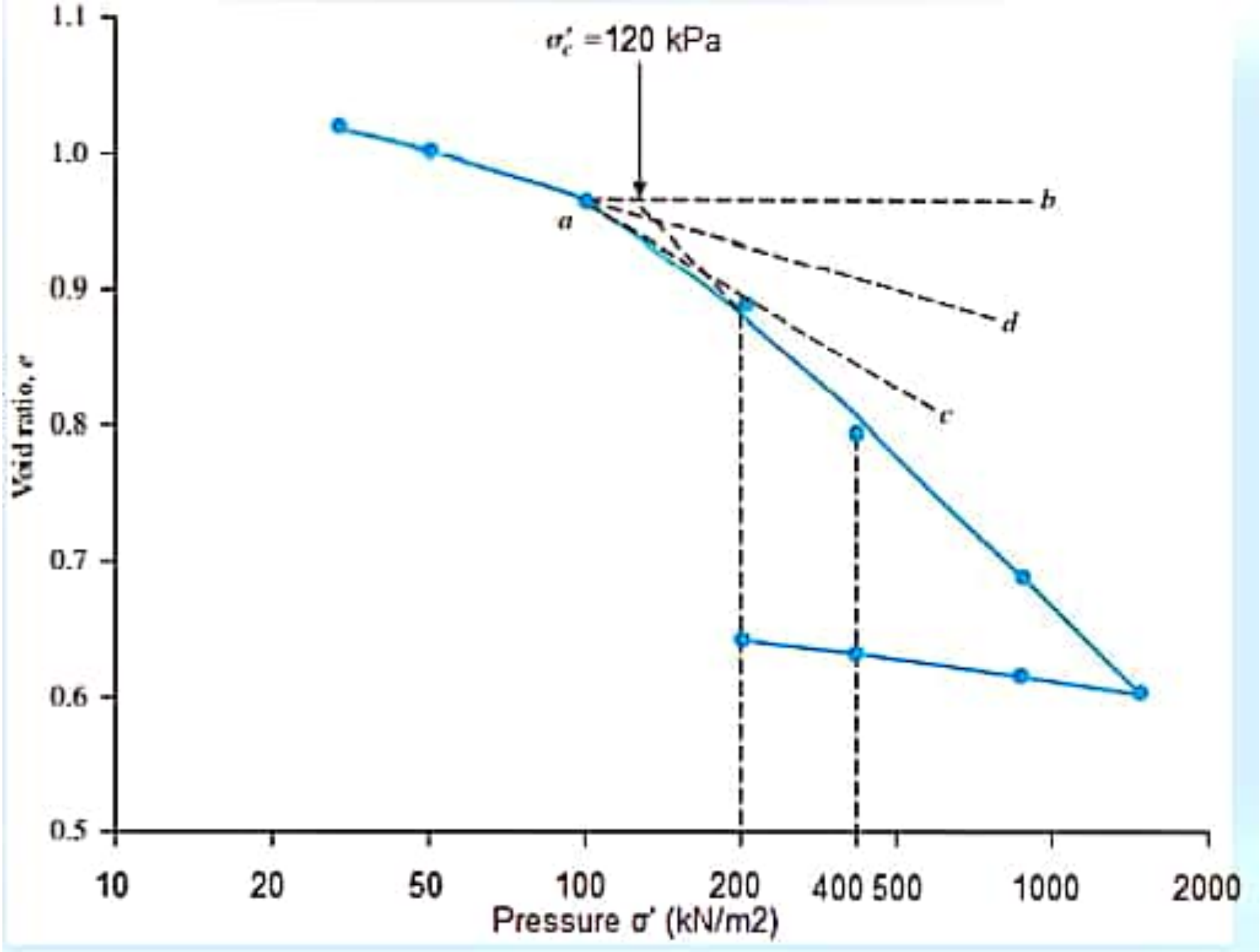
①

Pressure (σ') kN/m^2	e (Void Ratio)
25	1.03
50	1.02
100	0.98
200	0.91
400	0.79
800	0.71
1600	0.62
800	0.635
400	0.655
200	0.67

a) Draw e-log σ' graph and determine the preconsolidation Pressure (σ_p' or σ_c')

b) calculate the compression index and the ratio of C_s/C_c

c) On the basis of the average e-log σ' plot, calculate the void ratio at 200 KPa.


σ'_c =120 kPa
a
b
c
d
Void ratio, e
1.1
1.0
0.9
0.8
0.7
0.6
0.5
10
20
50
100
200
400 500
1000
2000
Pressure σ' (kN/m2)

b) Slope of the compression curve is equal to C_c

$$C_c = \frac{e_1 - e_2}{\log\left(\frac{\sigma_2'}{\sigma_1'}\right)} \rightarrow \text{loading}$$

σ	e
200	0.91
400	0.79

$$C_c = \frac{0.91 - 0.79}{\log\left(\frac{400}{200}\right)} = 0.39$$

Slope of the swelling curve is equal to C_s

$$C_s = \frac{e_1 - e_2}{\log\left(\frac{\sigma_2'}{\sigma_1'}\right)} \rightarrow \text{unloading}$$

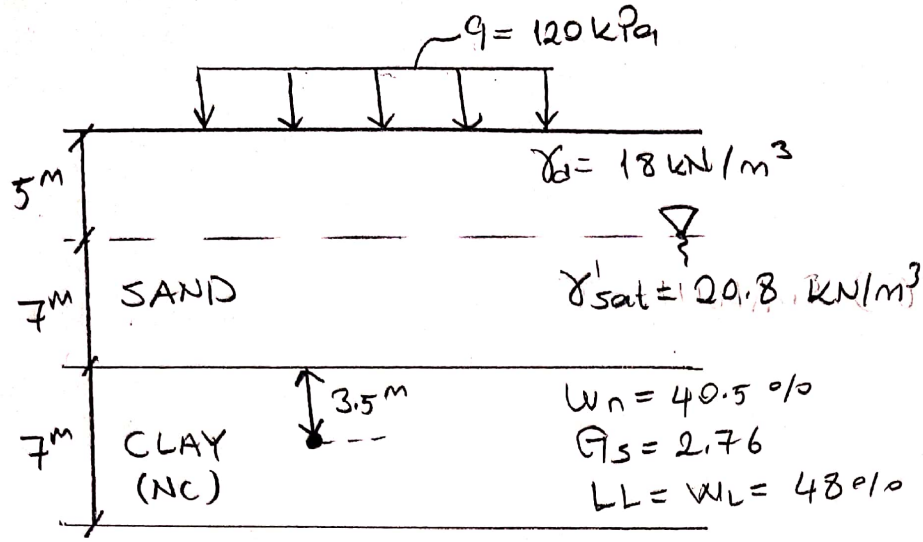
	σ	e
2)	400	0.655
1)	200	0.67

$$C_s = \frac{0.67 - 0.655}{\log\left(400/200\right)} = 0.049$$

$$\frac{C_s}{C_c} = \frac{0.049}{0.39} = 0.128$$

C) $$0.39 = \frac{0.91 - e}{\log\left(\frac{1200}{200}\right)} \Rightarrow e = 0.61$$

(2)



a) calculate the expected settlement of clay layer

Solution

$G_s \cdot w = S \cdot e$

$2.76 \cdot 0.405 = 1 \cdot e \Rightarrow e_o = 1.1178$

$\gamma_{sat,clay} = \left(\frac{G_s + e}{1+e}\right)\gamma_w = \left(\frac{2.76 + 1.1178}{1+1.1178}\right) \cdot 9.81$

$\gamma_{sat,clay} = 17.96 \text{ kN/m}^3$

- The vertical effective stress in the middle of the clay layer

$\sigma_o' = 18 \times 5 + (20.8 - 9.81) \times 7 + (17.96 - 9.81) \times 3.5$

$\sigma_o' = 195.5 \text{ kN/m}^2$

* In order to calculate settlement of the layer

C_c = compression Index.

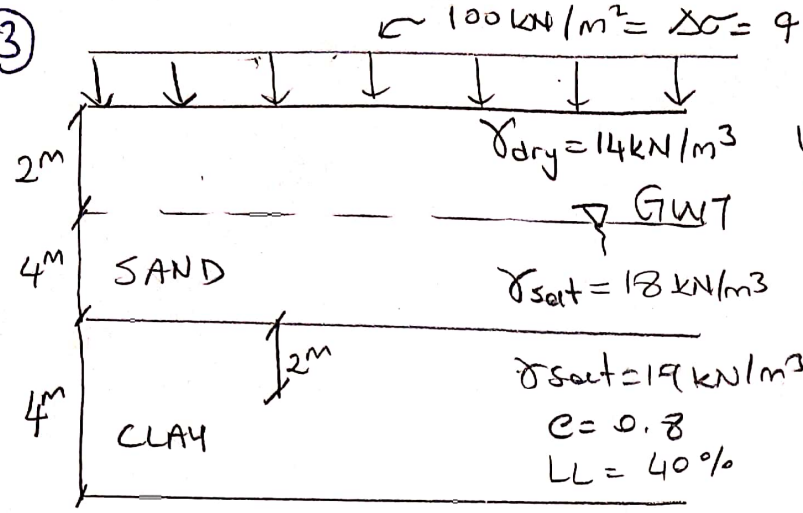
$C_c = 0.009 (w_L - 10) = 0.009 (48 - 10) = 0.34$ (w_L in %)

$S_c = \frac{C_c}{1+e_o} \cdot H \cdot \log\left(\frac{\sigma_o' + \Delta\sigma}{\sigma_o'}\right)$ $(\Delta\sigma = q = 120 \text{ kPa})$

$S_c = \frac{0.34}{1+1.1178} \cdot 700 \cdot \log\left(\frac{195.5 + 120}{195.5}\right) = 23.3 \text{ cm}.$

$\rightarrow S_c = 23.3 \text{ cm}$

(3)



What is the settlement of the clay layer caused by primary consolidation if!

a) The clay is normally consolidated

b) Over consolidated, $\sigma_p' = 190 kPa$

c) $\sigma_p' = 170'$

(assume the $C_s = \frac{1}{6} C_c$)

$\left(OCR = \frac{\sigma_p'}{\sigma'} \right)$

Solution

a) $\sigma_o' = 2 \times \gamma_{dry} + 4 \times (\gamma_{sat} - \gamma_w) + 2 \times (\gamma_{sat} - \gamma_w)$

$\sigma_o' = 2 \times 14 + 4 \times (18 - 9.81) + 2 \times (19 - 9.81)$

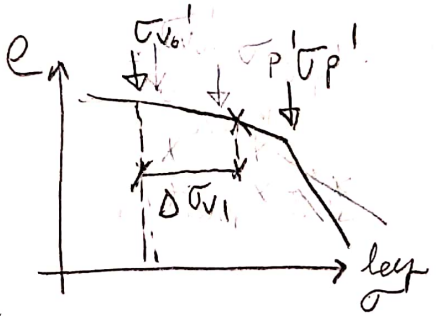
$\sigma_o' = 79.14 \ kN/m^2$

for normally consolidated clays.

$C_c = 0.009 (LL - 10) = 0.009 (40 - 10) = 0.27$

$$S_c = \frac{C_c}{1+e_o} . H . \log \left(\frac{\sigma_o' + \Delta\sigma}{\sigma_o'} \right)$$

$$= \frac{0.27}{1+0.8} . 4 . \log \left(\frac{79.14 + 100}{79.14} \right) = 21.3 \ cm$$



b) $\sigma_o' + \Delta\sigma' = 79.14 + 100 = 179.14$

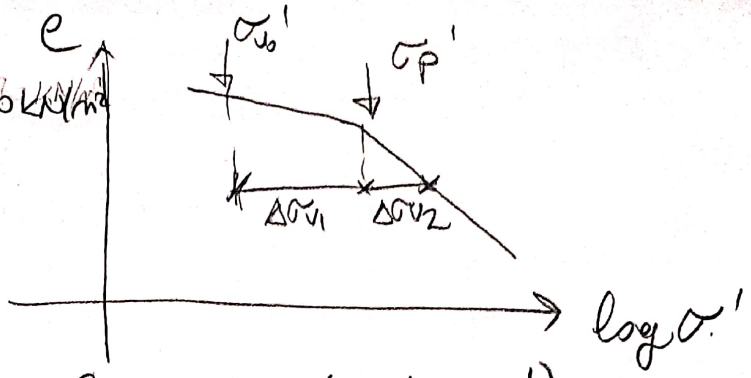
$\sigma_p' = 190 \ kN/m^2$

$\sigma_o' + \Delta\sigma < \sigma_p'$ → Use swelling coefficient.

$$S_c' = \frac{C_s}{1+e_o} . H . \log \left(\frac{\sigma_o' + \Delta\sigma'}{\sigma_o'} \right) \qquad C_s = \frac{0.27}{6} = 0.045$$

$$S_c = \frac{0.045}{1+0.8} . 4 . \log \left(\frac{79.14 + 100}{79.14} \right) = 3.6 \ cm$$

c) $\sigma_0' + \Delta\sigma' = 179.14\ kPa > \sigma_p' = 170\ kN/m^2$



$$S_c = \frac{C_s}{1+e_0} \cdot H \cdot \log\left(\frac{\sigma_p'}{\sigma_0'}\right) + \frac{C_c}{1+e_0} \cdot H \cdot \log\left(\frac{\sigma_0' + \Delta\sigma'}{\sigma_p'}\right)$$

$$S_c = \frac{0.045}{1+0.8} \cdot 4 \cdot \log\left(\frac{170}{79.14}\right) + \frac{0,27}{1+0.8} \cdot 4 \cdot \log\left(\frac{79.14+100}{170}\right)$$

$$S_c = 4.68\ cm$$